

END-TO-END DATA ANALYTICS REPORT

Climate Change Twitter Dataset (2006–2019)

1. INTRODUCTION

1.1 Background

This project analyzes global public discourse on climate change using a dataset of over 15 million tweets spanning 13 years (2006–2019). The goal is to support a climate research organization in understanding how sentiment and stance have evolved over time and to identify topics and regions driving aggressive or divisive discourse

1.2 Objectives

- This analysis aims at;
- Understanding dataset structure and quality
- Cleaning and preparing data for analysis
- Transforming data into analysis-ready formats
- Performing descriptive and diagnostic analytics
- Building an interactive Power BI dashboard
- Providing actionable insights

1.3 Tools Used

PostgreSQL for Data storage, cleaning, transformation

Power BI for Visualization and dashboarding

2. DATA UNDERSTANDING AND PREPARATION

2.1 Dataset Overview

The dataset contains 15,789,411 rows

Variables include:

Timestamp creation date

Sentiment score

Tweeters stance

Topics discoursed

Latitude & longitude

Gender

Temperature average

Aggressiveness of tweets

2.2 Data Structure & Schema

The dataset contains a single table with both categorical and numerical variables.

Key variable types:

Numeric : sentiment, temperature_avg

Categorical: topic, stance, gender, aggressiveness

Temporal: created_date

Geospatial : latitude, longitude

2.3 Data Range Validation

Sentiment range: -0.9942 to 0.9917

Longitude range: -179.66 to 179.38

Latitude range: -90 to 85

These ranges confirm that values fall within expected real-world boundaries.

2.4 Data Quality Assessment

Missing Values:

Latitude & Longitude and Temperature_avg had missing rows of 10,481,873

Others columns had zero missing rows

Undefined Values:

Gender includes "undefined" = 586,875 rows

Duplicates:

No duplicate rows detected

Key Observations:

Large proportion of tweets lack geolocation , making up to 66.38% of total tweets without geolocation

Dataset remains valid for sentiment and topic analysis

Missing coordinates were handled without deletion

2.5 Data Cleaning Strategy

Instead of deleting data:

Missing coordinates → labeled as “unknown” region

3. DATA WRANGLING AND TRANSFORMATION

3.1 c_climate_dataset

A cleaned view (c_climate_dataset) was created to:

Standardize categorical variables. The category emails were trimmed and standardized to lower case

Create derived columns: year , month , region, sentiment category and time period column was derived

Handle missing values: missing values in the longitude, latitude and temp_avg column were handled by creating a regional_analysis view conditioned to when latitude and longitude is not null

3.2 Feature Engineering

Derived Columns:

Year : extracted from timestamp

Sentiment Label:

Positive : used percentile cont of 0.75

Negative : used percentile cont of 0.25

Neutral (otherwise)

Region : derived from coordinates with data. Where coordinates was null, region was laelled “unknown” . Other region that didn’t fall between the assumed coordinate range was labelled “others”. 33,011 rows were labelled as others, making about 0.6% Of rows with coordinates.

3.3 Data Structuring (Views Created)

To optimize analysis, two views were created:

Overall analysis view as c_climate_dataset

Regional analysis view

3.4 Data Handling Decisions

Full dataset retained (no unnecessary row deletion)

Missing values handled contextually

Views used instead of modifying raw data

4. DESCRIPTIVE ANALYTICS

4.1 Summary Statistics

Sentiment category Distribution:

Negative: 3,947,360

Positive: 3,952,791

Neutral: 7,889,260

1. Topic Summary

Topic analysis shows that climate-related discussions are concentrated around a few dominant themes, with some topics generating significantly higher tweet volumes than others. This indicates that public attention is focused on specific areas of climate discourse.

Average sentiment varies across topics, suggesting that certain subjects are viewed more positively while others attract more negative reactions. Additionally, sentiment distribution reveals that some topics exhibit a balanced mix of opinions, while others are more polarized, containing higher proportions of either positive or negative sentiment.

2. Regional Summary

Regional analysis indicates that tweet activity is unevenly distributed across geographic areas, with some regions contributing a significantly larger share of the total conversation.

Sentiment varies across regions, reflecting differences in perception and engagement with climate change issues globally. While some regions show relatively positive sentiment, others lean more negative.

Sentiment distribution across regions suggests varying levels of polarization, with certain regions exhibiting a more balanced mix of opinions and others showing stronger concentration in specific sentiment categories.

3. Gender Summary

Gender-based analysis shows that tweet participation is distributed across male, female, and undefined categories, with a portion of the dataset lacking clear gender identification.

Average sentiment across genders is relatively similar, indicating no extreme divergence in opinion between male and female users. This suggests that climate change discourse is broadly shared across genders.

Sentiment distribution further confirms that both male and female users contribute to positive, negative, and neutral discussions, with no single gender dominating a particular sentiment category.

4. Aggressiveness Summary

Aggressiveness analysis reveals that the majority of tweets are non-aggressive, indicating that most climate-related discussions occur in a relatively calm tone.

However, aggressive tweets exhibit a noticeably more negative average sentiment (-0.1016) compared to non-aggressive tweets (0.044), highlighting a strong relationship between aggressive language and negative emotional expression.

Distribution analysis shows that aggressive discourse is not evenly spread across all topics, with certain topics exhibiting a higher proportion of aggressive tweets. This suggests that specific areas of climate discussion are more contentious and emotionally charged.

FINAL OVERALL STATEMENT

Overall, descriptive analysis shows that climate change discourse is highly active but varies significantly across topics, regions, and user groups. While sentiment remains broadly balanced between positive and negative opinions, the presence of aggressive and polarized discussions in specific areas highlights the complexity of public engagement with climate issues.

4.2 Trends and Patterns

- There is a significant relationship between negative sentiment and aggression . analysis showed that tweets with negative sentiment show higher proportion of aggressiveness

Aggression Over Time:

Early: 31.61%

Mid: 32.92% (peak)

Late: 27.59% (decline)

Key Patterns Identified:

Aggression peaked mid-period before declining

Sentiment stabilized in later years

Certain topics such as politics consistently drive high aggression while pollution awareness drives the least

4.3 Initial Exploration of Key Variables

Sentiment values are well distributed across categories

Topics show concentration around key themes

Region variable includes significant “unknown” category

Aggressiveness skewed toward non-aggressive behavior

5. DIAGNOSTIC ANALYTICS

5.1 Sentiment and Aggression Relationship

Negative sentiment shows the highest aggression rates

Indicates emotional negativity drives aggressive discourse

5.2 Topic Drivers of Aggression

Certain topics exhibit significantly higher aggression

These topics are likely controversial or polarizing

5.3 Regional Differences

Regions differ in:

Sentiment levels

Aggression rates

Suggests geographic variation in climate discourse

5.4 Stance and Behaviour

Specific stances show higher aggression

Indicates belief systems influence tone of communication

5.5 Gender Influence

Male users show relatively higher aggression levels

Behaviour varies when combined with stance

5.6 Time-Based Explanation

Mid-period aggression spike linked to:

Increased engagement

Controversial discussions

Later decline suggests stabilization

5.7 Polarization Analysis

Topics with high sentiment variation show:

Strong disagreement

High divisiveness